INCH-POUND

MIL-C-70549 (AR)
AMENDMENT 5
22 July 1996
SUPERSEDING
AMENDMENT 4
14 July 1994

MILITARY SPECIFICATION

CARTRIDGE, 81MM, SMOKE, RED PHOSPHORUS
M819, WITH FUZE, MTSQ, M772
LOADING, ASSEMBLING AND PACKING

This amendment forms a part of MIL-C-70549 (AR), dated 5 September 1986, and is approved for use by the U.S. Army Armament Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1.2, Heading: Delete "DRAWINGS", and substitute: "PRODUCT AND PACKAGING DRAWINGS".

Under PRODUCT AND PACKAGING DRAWINGS: Delete:

"8796521 - Marking ... Containers.
8796522 - Marking ... Boxes."

Add:

"12944509 - Packing and Marking for Container, Ammo, PA156 and PA157; for Cartridge, 81mm."

After PRODUCT AND PACKAGING DRAWINGS: Add:

"INSPECTION EQUIPMENT DRAWINGS

6518071 - Alignment Gage.
6518072 - Alignment Check Gage."

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PAGE 4

3.4.6, Delete in its entirety and substitute:

"3.4.6 Fuze function: The fuze shall air burst when set in accordance with paragraph 4.5.5, Table III."

PAGE 5

4.3.1, Delete: "Body tube assembly 9327832 5".

Delete: "9362899" and substitute: "9362799".

Add the following under "Projectile loading assembly ... 9327838 ... 5*".

"Projectile loading assembly (when LAP is split between two facilities) ...9327838 ...50*".

In last line under "Quantity" delete: "5" and substitute: "50*".

After list add "*Same group of samples when LAP is split".

PAGE 6

TABLE I, Paragraph Reference column: Add "4.4.2.2" before "4.4.2.5" ".

Under Projectile Loading Assembly: Add:

"1/" next to "5" in "No. of sample units" column.

Under Projectile Loading Assembly ...Examination for defects: Add:

"Ballistic test ... 50 2/ ... 100% ... 3.2 ...4.5.5".

Under Cartridge, 81MM Smoke ... Examination for defects, in No. of sample units column: Delete:

"5" and substitute: "50".

Under Ballistic test, in No. of sample units column: Add:

"3/" next to "50".

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PAGE 6 (Cont'd)

Notes: Add:

"1/ Sample shall be 50 when LAP is split
2/ Required only when LAP is split
3/ Same units as in Note 2/ when LAP is split".

Delete: "9354333" and substitute: "9362799" where it appears.

Delete: "9362899" and substitute: "9362799"

Under ballistic test sample quantity: Delete: "5" and substitute: "50".

PAGE 8

4.4.2.1: Delete in its entirety.

PAGE 9

4.4.2.2., Major 105, Delete: "Piston plate missing or damaged" and substitute: "Piston plate assembly missing or foil punctured".

Add Majors 110, 111, 112 and 113 as follows:

"110 ...Presence of foreign matter between foil disc and piston plate or presence of contaminant on piston plate ... 0.40% ... 3.2 ... Visual/Manual

111 ...Foil punctured, wrinkled or not adhering fully to piston plate ... 0.40% ... 3.2 ... Visual

112 ...Concentricity of foil disc with piston plate outside diameter (OD) ... 0.40% ... 3.2 ... Gage

113 ...Presence of foreign matter in tail cone groove ... 0.40% ... 3.2 ... Visual".

PAGE 10

4.4.2.3, Major 105: delete "5" in "No. of sample units" column and delete "100%" in "AQL or 100%" column.

Note 1/: Delete in its entirety and substitute:

"1/ See 4.4.3.1".

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PAGE 12

4.4.2.5, Major 109: Delete: "0.40% ... 3.2 ... Visual" and substitute: "100% ... 3.2 ... Gage".

PAGE 13

4.4.2.5, Delete Major 112 in its entirety. and substitute:

"Major 112 ... Assembly fails to freely traverse alignment gage ... 1/ ... 3.2 ... Gage".

Notes: Add:

"1/. Applies to split LAP only. At start-up, 30 consecutive assemblies must successfully pass. Thereafter, one out of each 20 assemblies shall be checked. If any failures occur, the preceding 50 assemblies shall be checked. If any more failures occur, continue to screen groups of 50 until all pass."

PAGE 15

4.4.2.7, Next Higher Assembly: Delete: "9362899" and substitute: "9362799".

Major 106: Delete in its entirety and substitute:

"106 Inadequate torque applied to fin boom or fuze ... 1/ ... (-) ... 3.2 ... Torque test/4.5.2".

Note 1/: Add:

"and 5 assemblies per hour thereafter. Any failures shall cause rejection back to the last successful inspection."

PAGE 16

4.4.2.7: Delete "4.4.2.8" in paragraph block and substitute "4.4.2.7".

Next Higher Assembly: Delete: "9362899" and substitute: "9362799".

Major 109, "AQL" column: Delete: "100%".

Note 1/: Delete second sentence in its entirety and substitute:

"Repaint and recheck if the ring inspection exposes bare metal."

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PAGE 16 (Cont'd)

Major 110: Delete in its entirety.

Major 111: Delete: "Foil tape not centered on back of fuze" and substitute:

"Foil tape touches any part of fuze threads".

PAGE 16a

* 4.4.2.8: Add the following:

4.4.2.8	TITLE Support, Propelling Charge	SHEET 1 OF 1		DRAWING NUMBER 12977198
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 9362799 INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Width of opening	1/	3.2	Gage
102	Inside length	1/	3.2	Gage
103	Radius to outside surface of rib three (3) places	1/	3.2	Gage
104	Inside radius to support wall	1/	3.2	Gage
Minor				
201	Evidence of poor workmanship	Level V	3.6	Visual

NOTES:
1/ Sample three (3) parts from each mold cavity at the start of each mold run. All dimensions shall be as specified in the drawing prior to production run.

PAGE 17

4.4.2.9, Drawing number: Delete: "9354333" and substitute: "9362799".

Next higher assembly: Delete: "9362799".

Next higher assembly: Add: "/12944509" after "9362798".

Majors: Add the following:

"104 Propelling charge support missing or damaged ... 0.40% ... 3.2 ... Visual"

* Delete 105, 106, and 107 in their entirety.

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PAGE 18

4.4.2.10, Drawing Number: Delete: "9362899" and substitute: "9362799".

Drawing number: Delete: "/8796521".

Next higher assembly: Delete: "9354616" and substitute "9362798/12944509".

Major: Add:

"105 ... Tab too small ... - ... 0.40% ... 3.2 ...Gage".

PAGE 19

4.4.2.11, Title: Delete in its entirety and substitute:

"Box, Packing for Container, Ammunition Fiber for Cartridges, 81MM, M819/M853, prior to closing".

Drawing number: Delete: "9354616" and substitute: "9362798".

Next higher assembly: Delete: "9362798/8796522".

4.4.2.11, Major 102: Delete in its entirety.

Add Minors 203 and 204 as follows:

"203 Plastic sheet not present in every sixth box ... (-) ... 0.65% ... 3.2 ... Visual

204 Box containing plastic sheet not appropriately marked ... (-) ... 0.65% ... 3.2 ... Visual".

Add Minors 205 and 206 as follows:

"205 Graphical firing scale not present in every third box ... (-) ... 0.65% ... 3.2 ... Visual

206 Box containing graphical firing scale not appropriately marked ... (-) ... 0.65% ... 3.2 ... Visual".

PAGE 20

Title: Delete: "Wirebound" and substitute: "Wooden" where it appears.

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PAGE 20a

4.4.2.13: Add the following:

4.4.2.13	TITLE Packing and Marking for Container, Ammo, PA156 and PA157 for Cartridge, 81mm (prior to closing)	SHEET 1 OF 1		DRAWING NUMBER 12944509
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY / INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Containers loose or improperly packed in metal can	0.40%	3.2	Visual/Manual
102	Support assembly damaged, missing, or improperly placed	0.40%	3.2	Visual
Minor				
201	Circular firing scale not present in every third metal box	0.65%	3.2	Visual
202	Ammo box containing circular firing scale not appropriately marked	0.65%	3.2	Visual
203	Plastic sheet not present in every sixth box	0.65%	3.2	Visual
204	Ammo box containing plastic sheet not appropriately marked	0.65%	3.2	Visual
205	Evidence of poor workmanship	0.65%	3.2	Visual
NOTES:				

PAGE 20b

4.4.2.14: Add the following:

4.4.2.14	TITLE Packing and Marking for Container, Ammo, PA156 and PA157 for Cartridge, 81mm (after closing)	SHEET 1 OF 1		DRAWING NUMBER 12944509
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY / INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Contents of box loose	0.40%	3.2	Manual
102	Cover assembly not secure	0.40%	3.2	Manual
Minor				
201	Marking missing, misleading or unidentifiable	0.65%	3.2	Visual
202	Evidence of poor workmanship	0.65%	3.2	Visual
NOTES:				

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Under "Precaution..." add new paragraph 4.4.3.1 as follows:

"4.4.3.1 Tack weld pull test. Ten consecutive assemblies shall pass this test whenever equipment is started. Five (5) shall be tested each hour thereafter and ten (10) whenever tooling is changed or equipment is adjusted. Failure of any sample shall cause rejection of all obturators welded since the last successful test. See 4.5.4."

4.4.5, Table II: Delete "Fuze non-function ... 2" and substitute:

"Fuze non-functions, functions upon ground impact or fails to separate tail cone from body tube assembly ...1 ...2 (See Note 1)".

Delete: "Major ... Fuze functioning time (See Note 1) ... 2 ... 3".

Note 1: Delete in its entirety and substitute:

"Note 1/: If two (2) failures occur, another sample of 50 shall be fired. If two (2) or more additional failures occur the lot shall be rejected. If three (3) or more failures occur during the first 50 fired, the lot shall be rejected."

4.5.2: Delete: "4.2.3.4" and substitute: "4.4.2.7".

Add to the end of the last sentence:

"and shall cause rejection of all cartridges produced since the last successful test."

4.5.5: Delete full sentences 5, 6 and 7 ("However ... meters") in their entirety and substitute:

"Observation shall be made for compliance with the requirements specified for security of parts, air burst, horizontal range and smoke production. Horizontal range shall be reported to the point on the ground under the position of fuze functioning and shall be measured to the nearest five (5) meters."

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Delete: "The cartridges shall be conditioned at 70 plus or minus 2.5 degrees Fahrenheit for a minimum of 4 hours prior to testing and fired at this temperature." and substitute:

"The cartridges shall be conditioned at 70 plus or minus 2.5 degrees Fahrenheit for a minimum of 24 hours prior to testing and shall be fired at this temperature."

After the last sentence: add:

"The cannon shall be dry swabbed after every tenth round fired, regardless of charge."

TABLE III: Delete in its entirety and substitute:

"TABLE III

FUZE SETTING (sec.)	CHARGE ZONE	ELEVATION (mils)
24.5	1	1459
46.0	4	1489"

5.1: Delete in its entirety and substitute:

"5.1 Unit packaging.

5.1.1 Level A. Packaging for cartridges shall be in accordance with the requirements of drawing 9362799. Prior to taping, the felt spacers shall cause a gap of approximately 1/8 inch between the cap and tube of the fiber container when unrestrained.

5.1.2 Level B. Same as Level A.

5.1.3 Level C. Not applicable."

5.2: Add new paragraph as follows::

"5.2 Packing.

5.2.1 Level A. Cartridges shall be packed in accordance with the requirements of drawing 9362798 or 12944509, as applicable.

5.2.2 Level B. Same as Level A.

5.2.3 Level C. Not applicable."

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5.3 Add new paragraph as follows:

"5.3 Marking.

5.3.1 Level A. Packaged cartridges shall be marked in accordance with the requirements of drawing 9362799. Packed cartridges shall be marked with the requirements of drawing 9362798-1 or 12944509, as applicable.

5.3.2 Level B. Same as Level A.

5.3.3 Level C. Not applicable."

* 6.4: Delete: "SMCAR-QAT-I, Dover, NJ 07801-5001" and substitute:

"AMSTA-AR-QAA-C, Picatinny Arsenal, NJ 07806-5000".

* 6.5: Delete: "SMCAR-QAT-B/SMCAR-FSA-MM, Picatinny Arsenal, NJ 07806-5000" and substitute:

"AMSTA-AR-QAA-C/AMSTA-AR-FSA-M, Picatinny Arsenal, NJ 07806-5000".

6.6: Delete in its entirety and substitute:

"6.6 Alternate inspection provisions. Alternative inspection procedures, methods or equipment (including tool control inspection, statistical process control, substitute sampling plans, etc.) may be used by the contractor when he provides, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative inspection procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Procuring Contracting Officer for evaluation and approval by the government. When required, the contractor shall demonstrate that the effectiveness of the proposed alternative(s) is equal to, or better than, the quality assurance provisions herein. In cases of dispute as to whether certain procedures of the contractor's proposed alternative(s) provides equal assurance, the provisions of this specification shall apply."

On Form 1426, Delete: "M19" and substitute: "M819" where it appears.

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